从 Cisco 交换机迁移到 Cisco Nexus 92300YC 交换机

ONTAP Systems Switches

NetApp
January 31, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/ontap-systems-switches/switch-cisco-92300/migrate-to-92300yc-overview.html on January 31, 2022. Always check docs.netapp.com for the latest.

目录

从 Cisco 交换机迁移到 Cisco Nexus 92300YC 交换机 1
从 Cisco 交换机迁移到 Cisco Nexus 92300YC 交换机 1
如何从 Cisco 交换机迁移到 Cisco Nexus 92300YC 交换机 1

从 Cisco 交换机迁移到 Cisco Nexus 92300YC 交换机

从 Cisco 交换机迁移到 Cisco Nexus 92300YC 交换机

在将某些旧的 Cisco Nexus 集群交换机更换为 Cisco Nexus 92300YC 集群交换机时，您必须了解某些配置信息，端口连接和布线要求。

- 支持以下集群交换机：
  - Nexus 92300YC
  - Nexus 5596UP
  - Nexus 5020
  - Nexus 5010
- 集群交换机使用以下端口连接到节点：
  - 端口 E1/1-48 （10/25 GbE）， E1/49-64 （40/100 GbE）： Nexus 92300YC
  - 端口 E1/1-40 （10 GbE）： Nexus 5596UP
  - 端口 E1/1-32 （10 GbE）： Nexus 5020
  - 端口 E1/1-12， E2/1-6 （10 GbE）：具有扩展模块的 Nexus 5010
- 集群交换机使用以下交换机间链路（ISL）端口：
  - 端口 E1/6566 （100 GbE）： Nexus 92300YC
  - 端口 E1/41-48 （10 GbE）： Nexus 5596UP
  - 端口 E1/33-40 （10 GbE）： Nexus 5020
  - 端口 E1/13-20 （10 GbE）： Nexus 5010
- *cluster* Hardware Universe 包含有关所有集群交换机支持的布线的信息。
- 您已将 Nexus 92300YC 交换机上的某些端口配置为以 10 GbE 或 40 GbE 运行。
- 您已规划，迁移并记录从节点到 Nexus 92300YC 集群交换机的 10 GbE 和 40 GbE 连接。
- 此操作步骤中支持的 ONTAP 和 NX-OS 版本位于 _Cisco 以太网交换机 _ 页面上。

  "Cisco 以太网交换机"

> 迁移完成后，您可能需要安装所需的配置文件，以支持 923300 YC 集群交换机的集群交换机运行状况监控器（Cluster Switch Health Monitor， CSHM）。请参见中的 *Installing the Cluster Switch Health Monitor （CSHM） configuration file for 923300 YC switchs* "设置" 指南

如何从 Cisco 交换机迁移到 Cisco Nexus 92300YC 交换机

您可以将 ONTAP 集群中较早的 Cisco 集群交换机无中断迁移到 Cisco Nexus 92300YC 集群网络交换机。

关于此任务

- 现有集群必须已正确设置并正常运行。
- 所有集群端口都必须处于 up 状态，以确保无中断运行。
- Nexus 92300YC 集群交换机必须在安装的正确版本的 NX-OS 下进行配置并运行，并应用参考配置文件（Reference Configuration File ， RCF ）。
- 现有集群网络配置必须满足以下要求：
  - 使用两个旧版 Cisco 交换机的冗余且功能完备的 NetApp 集群。
  - 管理连接以及对旧 Cisco 交换机和新交换机的控制台访问。
  - 所有与集群 LIF 处于 up 状态的集群 LIF 均位于其主端口上。
  - ISL 端口已启用，并已在旧的 Cisco 交换机之间以及新交换机之间进行布线。

此操作步骤中的示例使用以下交换机和节点命名：

- 现有的 Cisco Nexus 5596UP 集群交换机为 C1 和 C2 。
- 新的 Nexus 92300YC 集群交换机为 CS1 和 CS2 。
- 节点为 node1 和 node2 。
- 集群 LIF 分别是节点 1 上的 node1_clus1 和 node1_clus2 以及节点 2 上的 node2_clus1 和 node2_clus2 。
- 交换机 C2 首先由交换机 CS2 取代，然后交换机 C1 由交换机 CS1 取代。
  - 临时 ISL 基于将 C1 连接到 CS1 的 CS1 构建。
  - 然后，节点和 C2 之间的布线将从 C2 断开，并重新连接到 CS2 。
  - 然后，节点和 C1 之间的布线将从 C1 断开，并重新连接到 CS1 。
  - 然后删除 C1 和 CS1 之间的临时 ISL 。

步骤

1. 将权限级别更改为高级，在系统提示您继续时输入 * y * ：

   ```
   set -privilege advanced
   ```

   此时将显示高级提示符（ * > ）。

2. 如果在此集群上启用了 AutoSupport ，则通过调用 AutoSupport 消息禁止自动创建案例：

   ```
   ssystem node AutoSupport invoke -node * -type all -message MAINT=xh
   ```

   其中 x 是维护时段的持续时间，以小时为单位。

   > AutoSupport 消息会通知技术支持此维护任务，以便在维护窗口期间禁止自动创建案例。

   以下命令将禁止自动创建案例 2 小时：

```
cluster1::*> **system node autosupport invoke -node * -type all -message
MAINT=2h**
```

3. 验证是否已在所有集群 LIF 上启用自动还原：

   `network interface show - vserver cluster -fields auto-revert`

```
cluster1::*> **network interface show -vserver Cluster -fields auto-
revert**

          Logical
Vserver   Interface     Auto-revert
--------- ------------- ------------
Cluster
          node1_clus1   true
          node1_clus2   true
          node2_clus1   true
          node2_clus2   true

4 entries were displayed.
```

4. 确定每个集群接口的管理或运行状态：

   对于 `Link` ，每个端口均应显示 up ；对于 `Health Status` ，每个端口均应显示 Healthy 。

   a. 显示网络端口属性：

      `network port show -ipspace cluster`

```
cluster1::*> **network port show -ipspace Cluster**

Node: node1

Ignore
                                                      Speed(Mbps) Health
Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper  Status
Status
--------- ------------ ---------------- ---- ---- -----------
-------- ------
e0a       Cluster      Cluster          up   9000  auto/10000 healthy
false
e0b       Cluster      Cluster          up   9000  auto/10000 healthy
false

Node: node2

Ignore
                                                      Speed(Mbps) Health
Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper  Status
Status
--------- ------------ ---------------- ---- ---- -----------
-------- ------
e0a       Cluster      Cluster          up   9000  auto/10000 healthy
false
e0b       Cluster      Cluster          up   9000  auto/10000 healthy
false

4 entries were displayed.
```

b. 显示有关逻辑接口及其指定主节点的信息：

   `network interface show -vserver cluster`

   对于 S状态管理 `/Oper` ，每个 LIF 都应显示 up/up ；对于 ，每个 `LIF` 都应显示 `true` ，而对于 ，则显示 Home 。

```
cluster1::*> **network interface show -vserver Cluster**

            Logical      Status     Network            Current
Current Is
Vserver     Interface    Admin/Oper Address/Mask       Node
Port    Home
----------- -----------  ---------- ------------------ -------------
------- ----
Cluster
            node1_clus1  up/up      169.254.209.69/16  node1         e0a
true
            node1_clus2  up/up      169.254.49.125/16  node1         e0b
true
            node2_clus1  up/up      169.254.47.194/16  node2         e0a
true
            node2_clus2  up/up      169.254.19.183/16  node2         e0b
true

4 entries were displayed.
```

5. 从节点的角度来看，每个节点上的集群端口均使用命令以以下方式连接到现有集群交换机：

   `network device-discovery show -protocol cdp`

```
cluster1::*> **network device-discovery show -protocol cdp**
Node/       Local  Discovered
Protocol    Port   Device (LLDP: ChassisID)  Interface         Platform
----------- ------ ------------------------- ----------------
----------------
node2      /cdp
            e0a    c1                        0/2               N5K-
C5596UP
            e0b    c2                        0/2               N5K-
C5596UP
node1      /cdp
            e0a    c1                        0/1               N5K-
C5596UP
            e0b    c2                        0/1               N5K-
C5596UP

4 entries were displayed.
```

6. 从交换机的角度来看，集群端口和交换机使用命令以以下方式进行连接：

s如何使用 cdp 邻居

```
c1# **show cdp neighbors**

Capability Codes: R - Router, T - Trans-Bridge, B - Source-Route-Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater,
                  V - VoIP-Phone, D - Remotely-Managed-Device,
                  s - Supports-STP-Dispute


Device-ID             Local Intrfce Hldtme Capability  Platform
Port ID
node1               Eth1/1         124    H         FAS2750
e0a
node2               Eth1/2         124    H         FAS2750
e0a
c2(FOX2025GEFC)     Eth1/41        179    S I s     N5K-C5596UP
Eth1/41

c2(FOX2025GEFC)     Eth1/42        175    S I s     N5K-C5596UP
Eth1/42

c2(FOX2025GEFC)     Eth1/43        179    S I s     N5K-C5596UP
Eth1/43

c2(FOX2025GEFC)     Eth1/44        175    S I s     N5K-C5596UP
Eth1/44

c2(FOX2025GEFC)     Eth1/45        179    S I s     N5K-C5596UP
Eth1/45

c2(FOX2025GEFC)     Eth1/46        179    S I s     N5K-C5596UP
Eth1/46

c2(FOX2025GEFC)     Eth1/47        175    S I s     N5K-C5596UP
Eth1/47

c2(FOX2025GEFC)     Eth1/48        179    S I s     N5K-C5596UP
Eth1/48

Total entries displayed: 10


c2# **show cdp neighbors**

Capability Codes: R - Router, T - Trans-Bridge, B - Source-Route-Bridge
```

```
                   S - Switch, H - Host, I - IGMP, r - Repeater,
                   V - VoIP-Phone, D - Remotely-Managed-Device,
                   s - Supports-STP-Dispute


Device-ID             Local Intrfce Hldtme Capability  Platform
Port ID
node1               Eth1/1         124    H         FAS2750
e0b
node2               Eth1/2         124    H         FAS2750
e0b
c1(FOX2025GEEX)     Eth1/41        175    S I s     N5K-C5596UP
Eth1/41

c1(FOX2025GEEX)     Eth1/42        175    S I s     N5K-C5596UP
Eth1/42

c1(FOX2025GEEX)     Eth1/43        175    S I s     N5K-C5596UP
Eth1/43

c1(FOX2025GEEX)     Eth1/44        175    S I s     N5K-C5596UP
Eth1/44

c1(FOX2025GEEX)     Eth1/45        175    S I s     N5K-C5596UP
Eth1/45

c1(FOX2025GEEX)     Eth1/46        175    S I s     N5K-C5596UP
Eth1/46

c1(FOX2025GEEX)     Eth1/47        176    S I s     N5K-C5596UP
Eth1/47

c1(FOX2025GEEX)     Eth1/48        176    S I s     N5K-C5596UP
Eth1/48
```

7. 使用命令确保集群网络具有完全连接：

```
cluster ping-cluster -node node-name
```

```
cluster1::*> **cluster ping-cluster -node node2**
Host is node2
Getting addresses from network interface table...
Cluster node1_clus1 169.254.209.69 node1     e0a
Cluster node1_clus2 169.254.49.125 node1     e0b
Cluster node2_clus1 169.254.47.194 node2     e0a
Cluster node2_clus2 169.254.19.183 node2     e0b
Local = 169.254.47.194 169.254.19.183
Remote = 169.254.209.69 169.254.49.125
Cluster Vserver Id = 4294967293
Ping status:
....
Basic connectivity succeeds on 4 path(s)
Basic connectivity fails on 0 path(s)
................
Detected 9000 byte MTU on 4 path(s):
    Local 169.254.19.183 to Remote 169.254.209.69
    Local 169.254.19.183 to Remote 169.254.49.125
    Local 169.254.47.194 to Remote 169.254.209.69
    Local 169.254.47.194 to Remote 169.254.49.125
Larger than PMTU communication succeeds on 4 path(s)
RPC status:
2 paths up, 0 paths down (tcp check)
2 paths up, 0 paths down (udp check)
```

8. 在 C1 和 CS1 之间的端口 E1/41-48 上的 cs1上 配置临时 ISL 。

   以下示例显示了如何在 C1 和 CS1 上配置新 ISL：

```
cs1# **configure**
Enter configuration commands, one per line. End with CNTL/Z.
cs1(config)# **interface e1/41-48**
cs1(config-if-range)# **description temporary ISL between Nexus 5596UP
and Nexus 92300YC**
cs1(config-if-range)# **no lldp transmit**
cs1(config-if-range)# **no lldp receive**
cs1(config-if-range)# **switchport mode trunk**
cs1(config-if-range)# **no spanning-tree bpduguard enable**
cs1(config-if-range)# **channel-group 101 mode active**
cs1(config-if-range)# **exit**
cs1(config)# **interface port-channel 101**
cs1(config-if)# **switchport mode trunk**
cs1(config-if)# **spanning-tree port type network**
cs1(config-if)# **exit**
cs1(config)# **exit**
```

9. 从 C2 的端口 E1/41-48 中拔下 ISL 缆线，并将缆线连接到 CS1 上的端口 E1/41-48 。
10. 验证连接 C1 和 CS1 的 ISL 端口和端口通道是否正常运行：

    s如何执行端口通道摘要

    以下示例显示了用于验证 C1 和 CS1 上 ISL 端口是否正常运行的 Cisco show port-channel summary 命令：

```
c1# **show port-channel summary**
Flags:  D - Down        P - Up in port-channel (members)
        I - Individual  H - Hot-standby (LACP only)
        s - Suspended   r - Module-removed
        b - BFD Session Wait
        S - Switched    R - Routed
        U - Up (port-channel)
        p - Up in delay-lacp mode (member)
        M - Not in use. Min-links not met
-------------------------------------------------------------------------
--------
Group Port-       Type     Protocol  Member Ports
      Channel
-------------------------------------------------------------------------
--------
1     Po1(SU)     Eth      LACP      Eth1/41(P)   Eth1/42(P)
Eth1/43(P)
                                     Eth1/44(P)   Eth1/45(P)
Eth1/46(P)
                                     Eth1/47(P)   Eth1/48(P)


cs1# **show port-channel summary**
Flags:  D - Down        P - Up in port-channel (members)
        I - Individual  H - Hot-standby (LACP only)
        s - Suspended   r - Module-removed
        b - BFD Session Wait
        S - Switched    R - Routed
        U - Up (port-channel)
        p - Up in delay-lacp mode (member)
        M - Not in use. Min-links not met
-------------------------------------------------------------------------
--------
Group Port-       Type     Protocol  Member Ports
      Channel
-------------------------------------------------------------------------
--------
1     Po1(SU)     Eth      LACP      Eth1/65(P)   Eth1/66(P)
101   Po101(SU)   Eth      LACP      Eth1/41(P)   Eth1/42(P)
Eth1/43(P)
                                     Eth1/44(P)   Eth1/45(P)
Eth1/46(P)
                                     Eth1/47(P)   Eth1/48(P)
```

11. 对于 node1 ，请断开 C2 上 E1/1 的缆线连接，然后使用 Nexus 92300YC 支持的相应布线方式将缆线连接

到 CS2 上的 E1/1 。

12. 对于 node2 ，请断开 C2 上 E1/2 的缆线连接，然后使用 Nexus 92300YC 支持的相应布线方式将缆线连接到 CS2 上的 E1/2 。
13. 现在，从节点的角度来看，每个节点上的集群端口均以以下方式连接到集群交换机：

    ```
    network device-discovery show -protocol cdp
    ```

    ```
    cluster1::*> **network device-discovery show -protocol cdp**

    Node/       Local  Discovered
    Protocol    Port   Device (LLDP: ChassisID)  Interface         Platform
    ----------- ------ ------------------------- ----------------
    ----------------
    node2      /cdp
                e0a    c1                        0/2               N5K-
    C5596UP
                e0b    cs2                       0/2               N9K-
    C92300YC
    node1      /cdp
                e0a    c1                        0/1               N5K-
    C5596UP
                e0b    cs2                       0/1               N9K-
    C92300YC

    4 entries were displayed.
    ```

14. 对于 node1 ，请断开 C1 上 E1/1 的缆线连接，然后使用 Nexus 92300YC 支持的相应缆线将缆线连接到 CS1 上的 E1/1 。
15. 对于 node2 ，请断开 C1 上 E1/2 的缆线连接，然后使用 Nexus 92300YC 支持的相应缆线将缆线连接到 CS1 上的 E1/2 。
16. 现在，从节点的角度来看，每个节点上的集群端口均以以下方式连接到集群交换机：

    ```
    network device-discovery show -protocol cdp
    ```

```
cluster1::*> **network device-discovery show -protocol cdp**
Node/       Local  Discovered
Protocol    Port   Device (LLDP: ChassisID)  Interface         Platform
----------- ------ ------------------------- ----------------
----------------
node2      /cdp
            e0a    cs1                       0/2               N9K-
C92300YC
            e0b    cs2                       0/2               N9K-
C92300YC
node1      /cdp
            e0a    cs1                       0/1               N9K-
C92300YC
            e0b    cs2                       0/1               N9K-
C92300YC
4 entries were displayed.
```

17. 删除 CS1 和 C1 之间的临时 ISL 。

```
cs1(config)# **no interface port-channel 10**1
cs1(config)# **interface e1/41-48**
cs1(config-if-range)# **lldp transmit**
cs1(config-if-range)# **lldp receive**
cs1(config-if-range)# **no switchport mode trunk**
cs1(config-if-range)# **no channel-group**
cs1(config-if-range)# **description 10GbE Node Port**
cs1(config-if-range)# **spanning-tree bpduguard enable**
cs1(config-if-range)# **exit**
cs1(config)# **exit**
```

18. 验证集群的最终配置：

    `network port show -ipspace cluster`

    对于 `Link` ，每个端口均应显示 up ；对于 `Health Status` ，每个端口均应显示 Healthy 。

```
cluster1::*> **network port show -ipspace Cluster**

Node: node1

Ignore
                                                   Speed(Mbps) Health
Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper  Status
```

```
Status
--------- ------------ ---------------- ---- ---- ----------- --------
------
e0a       Cluster      Cluster          up   9000  auto/10000 healthy
false
e0b       Cluster      Cluster          up   9000  auto/10000 healthy
false

Node: node2

Ignore
                                                   Speed(Mbps) Health
Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper  Status
Status
--------- ------------ ---------------- ---- ---- ----------- --------
------
e0a       Cluster      Cluster          up   9000  auto/10000 healthy
false
e0b       Cluster      Cluster          up   9000  auto/10000 healthy
false

4 entries were displayed.


cluster1::*> **network interface show -vserver Cluster**

            Logical    Status     Network            Current
Current Is
Vserver     Interface  Admin/Oper Address/Mask       Node          Port
Home
----------- ---------- ---------- ------------------ -------------
------- ----
Cluster
            node1_clus1  up/up    169.254.209.69/16  node1         e0a
true
            node1_clus2  up/up    169.254.49.125/16  node1         e0b
true
            node2_clus1  up/up    169.254.47.194/16  node2         e0a
true
            node2_clus2  up/up    169.254.19.183/16  node2         e0b
true

4 entries were displayed.
```

```
cluster1::*> **network device-discovery show -protocol cdp**

Node/       Local  Discovered
Protocol    Port   Device (LLDP: ChassisID)  Interface         Platform
----------- ------ ------------------------- ----------------
----------------
node2      /cdp
            e0a    cs1                       0/2               N9K-
C92300YC
            e0b    cs2                       0/2               N9K-
C92300YC
node1      /cdp
            e0a    cs1                       0/1               N9K-
C92300YC
            e0b    cs2                       0/1               N9K-
C92300YC

4 entries were displayed.


cs1# **show cdp neighbors**

Capability Codes: R - Router, T - Trans-Bridge, B - Source-Route-Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater,
                  V - VoIP-Phone, D - Remotely-Managed-Device,
                  s - Supports-STP-Dispute

Device-ID          Local Intrfce  Hldtme Capability  Platform      Port
ID
node1               Eth1/1         124    H         FAS2750
e0a
node2               Eth1/2         124    H         FAS2750
e0a
cs2(FDO220329V5)    Eth1/65        179    R S I s   N9K-C92300YC
Eth1/65
cs2(FDO220329V5)    Eth1/66        179    R S I s   N9K-C92300YC
Eth1/66


cs2# **show cdp neighbors**

Capability Codes: R - Router, T - Trans-Bridge, B - Source-Route-Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater,
                  V - VoIP-Phone, D - Remotely-Managed-Device,
                  s - Supports-STP-Dispute

Device-ID          Local Intrfce  Hldtme Capability  Platform      Port
```

```
ID
node1                Eth1/1         124    H         FAS2750
e0b
node2                Eth1/2         124    H         FAS2750
e0b
cs1(FDO220329KU)
                     Eth1/65        179    R S I s   N9K-C92300YC
Eth1/65
cs1(FDO220329KU)
                     Eth1/66        179    R S I s   N9K-C92300YC
Eth1/66

Total entries displayed: 4
```

19. 确保集群网络具有完全连接：

```
cluster ping-cluster -node node-name
```

```
cluster1::*> **set -priv advanced**

Warning: These advanced commands are potentially dangerous; use them
only when
         directed to do so by NetApp personnel.
Do you want to continue? {y|n}: **y**

cluster1::*> **cluster ping-cluster -node node2**
Host is node2
Getting addresses from network interface table...
Cluster node1_clus1 169.254.209.69 node1     e0a
Cluster node1_clus2 169.254.49.125 node1     e0b
Cluster node2_clus1 169.254.47.194 node2     e0a
Cluster node2_clus2 169.254.19.183 node2     e0b
Local = 169.254.47.194 169.254.19.183
Remote = 169.254.209.69 169.254.49.125
Cluster Vserver Id = 4294967293
Ping status:
....
Basic connectivity succeeds on 4 path(s)
Basic connectivity fails on 0 path(s)
................
Detected 9000 byte MTU on 4 path(s):
    Local 169.254.19.183 to Remote 169.254.209.69
    Local 169.254.19.183 to Remote 169.254.49.125
    Local 169.254.47.194 to Remote 169.254.209.69
    Local 169.254.47.194 to Remote 169.254.49.125
Larger than PMTU communication succeeds on 4 path(s)
RPC status:
2 paths up, 0 paths down (tcp check)
2 paths up, 0 paths down (udp check)


cluster1::*> **set -privilege admin**
cluster1::*>
```

20. 对于 ONTAP 9.4 及更高版本，启用集群交换机运行状况监控器日志收集功能以收集交换机相关的日志文件：

    `ssystem cluster-switch log setup-password system cluster-switch log enable-Collection`

```
cluster1::*> **system cluster-switch log setup-password**
Enter the switch name: <return>
The switch name entered is not recognized.
Choose from the following list:
NBS-NIC-NXYC-01
NBS-NIC-NXYC-02

cluster1::*> **system cluster-switch log setup-password**

Enter the switch name: **NBS-NIC-NXYC-01
**RSA key fingerprint is e5:8b:c6:dc:e2:18:18:09:36:63:d9:63:dd:03:d9:cc
Do you want to continue? {y|n}::[n] **y**

Enter the password: <enter switch password>
Enter the password again: <enter switch password>

cluster1::*> **system cluster-switch log setup-password**

Enter the switch name: **NBS-NIC-NXYC-02**
RSA key fingerprint is 57:49:86:a1:b9:80:6a:61:9a:86:8e:3c:e3:b7:1f:b1
Do you want to continue? {y|n}:: [n] **y**

Enter the password: <enter switch password>
Enter the password again: <enter switch password>

cluster1::*> **system cluster-switch log enable-collection**

Do you want to enable cluster log collection for all nodes in the
cluster?
{y|n}: [n] **y**

Enabling cluster switch log collection.

cluster1::*>
```

如果其中任何一个命令返回错误，请联系 NetApp 支持部门。

Copyright Information

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.